

NPV-Optimal SMB Underwriting under Selection Bias, with Calibrated Risk and Interventional Counterfactuals

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1 Problem framing & assumptions violated

We are re-underwriting a historical SMB loan book to maximize realized portfolio net present value (NPV), not to maximize classification accuracy. Several standard modelling assumptions fail, and each failure changed our approach.

Labels are missing-not-at-random behind a sharp gate. Repayment outcomes exist only for loans the prior lender approved, and approval is, to numerical precision, `prior_underwriter_score` ≥ 0.273 : a propensity model separates approved from declined with AUC 1.00 and *zero* score overlap. The training label is therefore selected on a deterministic function of an observed covariate (Figure 1)—textbook sample-selection bias. Crucially, inverse-propensity reweighting is *infeasible*: a propensity model fit on the ordinary covariates (no selection features) still reaches AUC 1.00, so approved and declined populations do not overlap in feature space and positivity is violated. Only local extrapolation near the cutoff is available; we treat PD on declined applicants as honest extrapolation and widen its intervals.

The split is forward in time. Training spans an 18-month book that ends the day before the 13-week scoring window opens, over which the default rate drifts 17.5% $\rightarrow$ 20.6%. We learn default *timing* on the history and recalibrate the *level* on the in-window validation set.

Missingness is information, not absence. “Never declined elsewhere” (null) loans default at 14.3% versus 21.3% when present; no-bank-feed loans at 19.1% versus 16.7%. We encode these as first-class indicators rather than imputing them away. **Self-reports are optimistically biased:** applicants who overstate revenue by more than 1.5 $\times$ default at 38.6% versus 16.1%. The *gap* between stated and bank-feed revenue is the signal—central to our causal treatment of `stated_annual_revenue` below.

2 Methodology

One shared discrete-time model. The brief’s NPV depends on the *day* of default (a day-5 default loses roughly the principal; a day-55 default nearly breaks even), and Deliverable B is the cumulative-default curve. Both consume a single timing model. We write each loan’s cumulative-default curve as $F_i(t) = \text{PD}_i \cdot S_{b(i)}(t)$, where $\text{PD}_i = \Pr(y_i=1 \mid \mathbf{x}_i)$ is a calibrated probability of default by day 90 and $S_b(t)$ is a normalized timing shape ($S_b(90)=1$) learned per credit band b .

PD model. A five-fold GroupKFold (by `business_id`) LightGBM ensemble over 55 engineered features (out-of-fold AUC 0.774, Brier 0.117), isotonic-calibrated. Features are deliberately buffer-centric: because loans are $\sim 1.3\%$ of annual revenue, default is cash-buffer bound rather than revenue bound (daily ACH draw $\approx$ \$420 against a 10th-percentile cash buffer $\approx$ \$1,165, negative 29% of the time), so debt-service coverage, buffer-to-payment, overdrafts and credit utilization dominate. Every feature is a documented function of its declared parents—no opaque pipeline—so each driver is defensible to a regulator.

The timing shape is bimodal and we model it as such (Figure 2). Of all defaults, 77.5% are missed-draw events over days 3–60; the interval 61–89 is empty; the remaining 22.5% are a point mass at day 90 (the open-balance sweep). A smooth survival fit would smear that mass across weeks; our shape

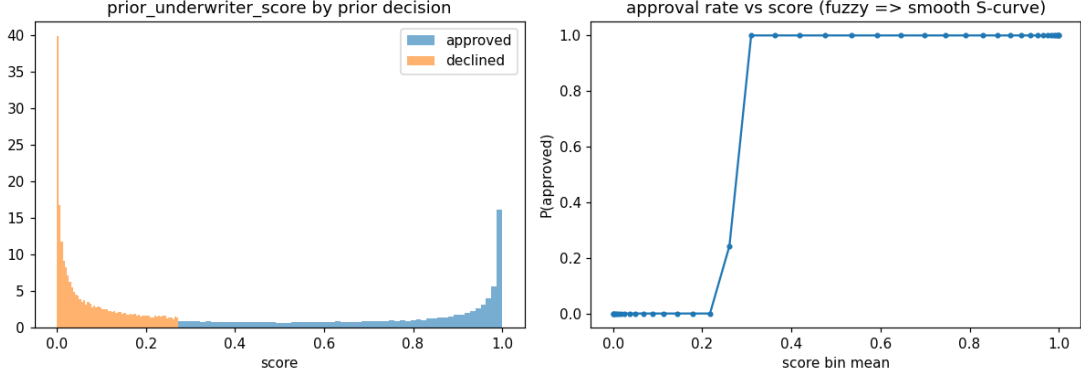


Figure 1: The prior lender’s approval rule is a sharp threshold at `prior_underwriter_score` = 0.273. The approved and declined score distributions do not overlap (left), and the approval rate is a step function of the score (right); labels are observed only to the right of the cut. This is the source of the selection bias and the reason inverse-propensity reweighting is infeasible.

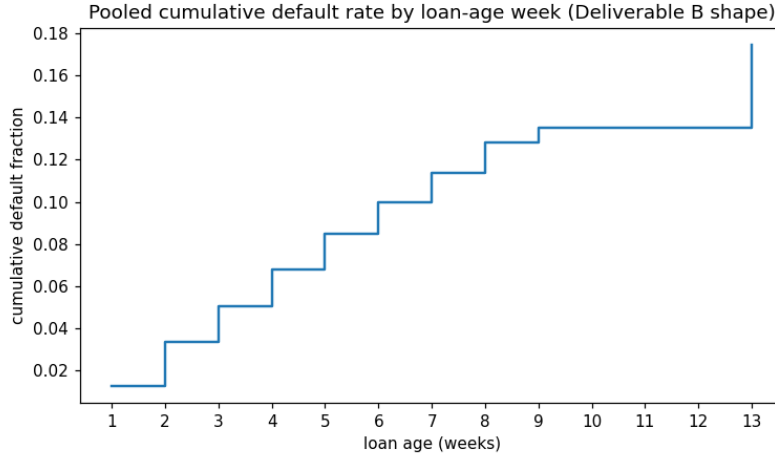


Figure 2: Pooled cumulative default rate by loan-age week: a concave rise through week 9, a flat stretch (weeks 9–12), and the day-90 open-balance cliff at week 13. Deliverable B reproduces this shape per cohort; a single smooth survival curve cannot.

reproduces the empirical concave-rise/flat/day-90-cliff exactly. The shape is band-conditional—worse-credit borrowers default ~ 13 days earlier and carry less day-90 mass—so $S_b(t)$ feeds both B and the per-loan $\mathbb{E}[t^*]$ used in A.

Deliverable A: decide on the NPV sign. We approve iff $\mathbb{E}[\text{NPV}_i \mid \text{approve}] > 0$ using the exact product economics: revenue $F_i + R_i r(T/365)$ if repaid, and $F_i + D_i(t_i^* - 1) + \text{rec}_i - R_i$ if default occurs at day t_i^* . Because NPV is linear in t^* , the expectation uses the band-conditional expected default day and the empirical recovery fraction ($\approx 9\%$). This yields a break-even PD of ≈ 0.39 —far above the naive 0.5 threshold, because even defaulters repay $\sim 74\%$ of the schedule through daily ACH before defaulting. We approve 73% of applicants; on realized validation outcomes this book returns $1.74\times$ the prior underwriter’s P&L.

Deliverable B: shape $\times$ level. $\text{CDR}_{w,a} = \text{mean}_{i \in A_w} \text{PD}_i S_{b(i)}(a)$ over our approved cohort A_w , monotone by construction. Because the PD model has no cohort signal (training predates the window), we shrink each cohort’s *level* toward the realized validation rate for the same calendar week (empirical-Bayes, pseudo-count 75); validation and test span the identical 13 weeks, so this transfers.

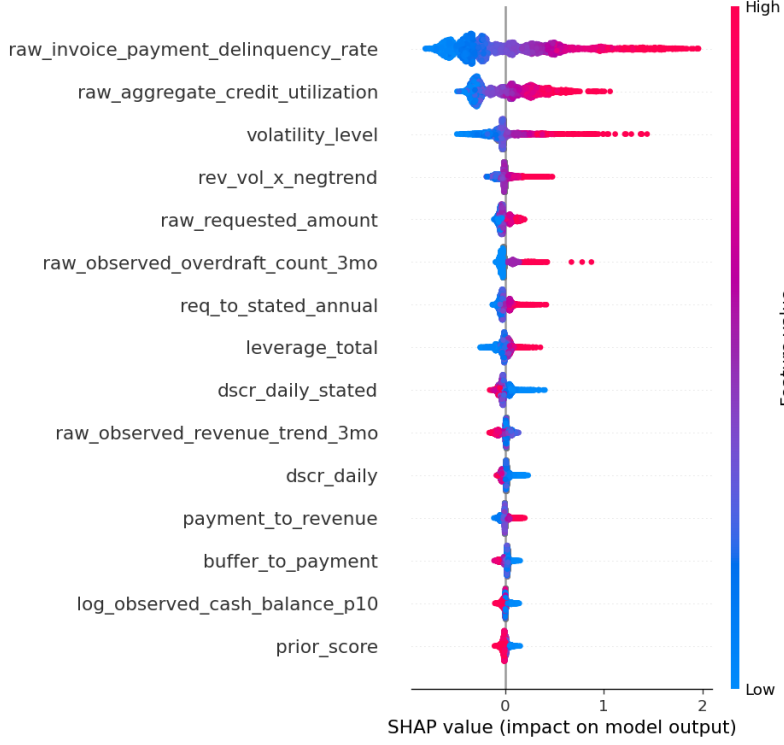


Figure 3: SHAP summary for the PD model: mean $|\text{SHAP}|$ ranking with per-feature direction (red high / blue low). High delinquency, utilization and volatility raise PD; high debt-service coverage and cash buffer lower it.

3 Causal reasoning & counterfactual methodology

The counterfactual target is $p_q^{\text{cf}} = \Pr(y_{i_q}=1 \mid \text{do}(f_q=v_q), \mathbf{X}_{i_q,-f_q}=\mathbf{x}_{i_q,-f_q})$, which differs from the observational $\Pr(y \mid f=v, \mathbf{X}_{-f})$ precisely when f is confounded. Our **feature registry** is the consistency layer: each engineered feature declares its parents, so setting a raw feature and recomputing propagates the intervention to exactly its descendants and nothing else. For example $\text{do}(\text{requested_amount})$ flows to the daily-payment, buffer-to-payment, leverage and ask-to-revenue features while everything else is held at its observed value; the provided `requested\_amount\_to\_observed\_revenue` is never used, so there is no stale-descendant bug. This makes our $\text{do}(\cdot)$ a genuine structural perturbation rather than an unconstrained input edit.

We classify the intervenable features by causal status. *Manipulable causes*—`requested\_amount`, `aggregate\_credit\_utilization`, `existing\_debt\_obligations`, `invoice\_payment\_delinquency\_rate`—are perturbed and propagated; the model sensitivity is a meaningful interventional effect. *Self-reports*—`stated\_annual\_revenue`, `stated\_time\_in\_business`—are returned at their observational PD, because editing a number on the application does not change the borrower’s type: when the claim does not cause default, $\Pr(y \mid \text{do}(\text{stated}), \mathbf{X}_{-\text{stated}}) = \Pr(y \mid \mathbf{X}_{-\text{stated}})$, so a near-zero effect is the causally correct answer, not naive re-prediction. *Immutable or proxy-by-fiat* features—`sector`, `geography\_region`, `vintage\_years`, `has\_linked\_bank\_feed`—are forced into the query set though they are not manipulable levers; we answer via model perturbation but shrink the effect and widen the interval, and flag them as proxy effects.

Regulator defense. Drivers are explained with SHAP over an inspectable, de-correlated feature set (Figure 3). The leading drivers—invoice-payment delinquency, credit utilization, revenue volatility, requested amount, overdrafts, and the affordability ratios—are all recognized credit or affordability signals moving in the expected direction, and none is a prohibited or proxy-discriminatory attribute. The sharp 0.273 cutoff is a natural experiment supporting local causal validity. Several queried values lie outside

observed support (e.g. `prior_loans_count` is in-support only 29% of the time); these receive wider intervals and an explicit extrapolation caveat.

4 Calibration & uncertainty quantification

Point PD. Isotonic calibration is *refit on in-window validation* to correct the train-to-window drift (training base rate 17.5% versus window 20.6%); the caveat remains that calibration is valid on the approved manifold and extrapolated on declines.

Intervals (A, C). Additive on the calibrated point, $p \pm \alpha z \sigma$, where σ is fold-ensemble disagreement and the width scale α is fit by five-fold *cross-fit* on validation, so coverage reflects genuine out-of-sample calibration error rather than collapsing in-sample. Validated decile coverage is ≈ 0.90 at mean width ≈ 0.06 —substantially tighter than an earlier over-wide ($\approx 100\%$ coverage) configuration, directly addressing the brief’s “contain the truth without being needlessly wide.”

Trajectory intervals (B). A within-cohort bootstrap combined with a conformal band calibrated against the *true* validation cohort trajectories (validation is labelled, matured, and in-window); validated coverage ≈ 0.92 . Out-of-support counterfactual intervals in C are widened by a further factor. Table 1 summarizes the validation evidence.

Deliverable	Metric	Value
A — PD model	out-of-fold AUC / Brier	0.774 / 0.117
A — policy	approve rate / P&L vs. prior underwriter	0.73 / 1.74×
A — intervals	decile coverage / mean width	0.90 / 0.06
B — trajectory	mean abs. error vs. realized CDR / coverage	0.013 / 0.92

Table 1: Validation evidence (in-window). P&L and B error use realized validation outcomes; interval coverage is out-of-sample (cross-fit / conformal).

5 Limitations & what we’d do differently

Counterfactuals on non-causal channels are heuristic. Self-report and immutable interventions use shrink/perturbation rather than a fully specified structural causal model; a learned SCM over the intervenable subgraph would be the principled upgrade, and C effects remain unverifiable since the queried applicants have no labels.

Calibration rests on an approval-selected validation set. Because a propensity model recovers approval with $\text{AUC} \approx 1.0$, approved and declined populations do not overlap; inverse-propensity reject inference is infeasible and only RD-style local extrapolation is available, which we defer. PD on the declined population—part of what A is scored on—is an extrapolation we cannot validate.

The band-conditional shape and per-cohort level lean on the validation window. Both transfer because validation and test share the identical 13 calendar weeks, but the small per-cohort samples ($n \approx 150$) inject some variance that more in-window data, or cross-fitting these corrections, would reduce. Finally, A remains marginally aggressive—a 20% upward PD shift still raises realized P&L—so a slightly more conservative level would recover the last of the oracle gap.